Synthetic PLC Skid

CP-1

Company profile	iec-default
Design rules	software defaults, not confirmed by the company's engineers

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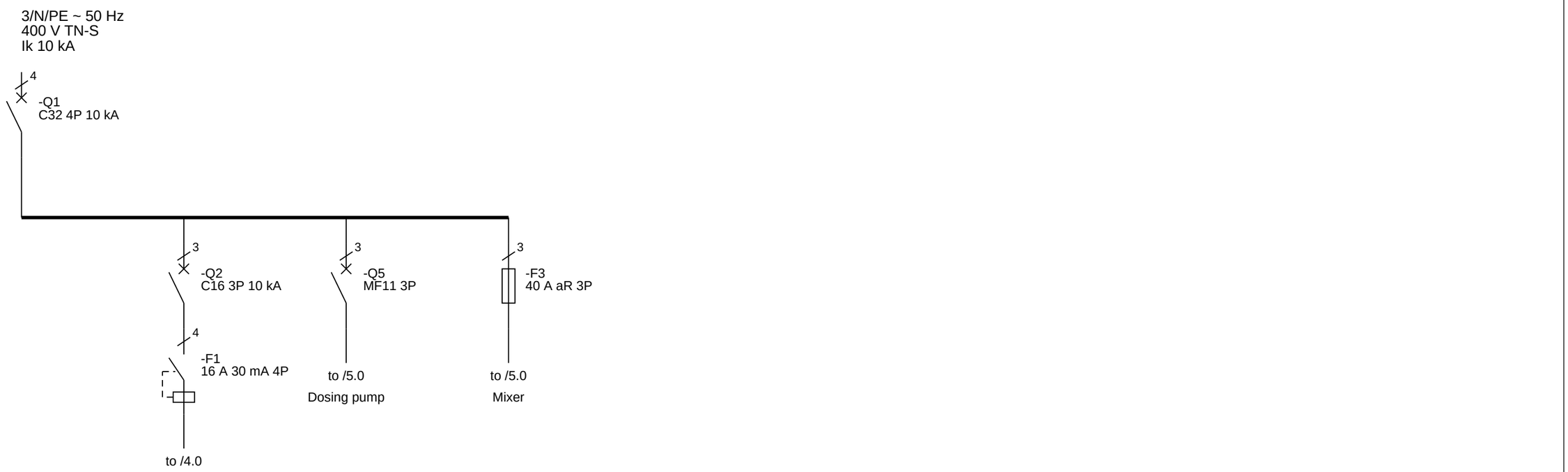

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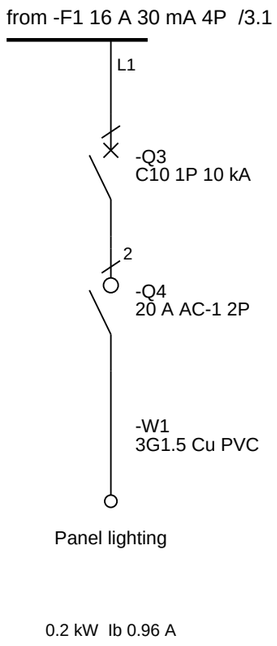

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	Synthetic PLC Skid		CP-1	**Distribution loads**	
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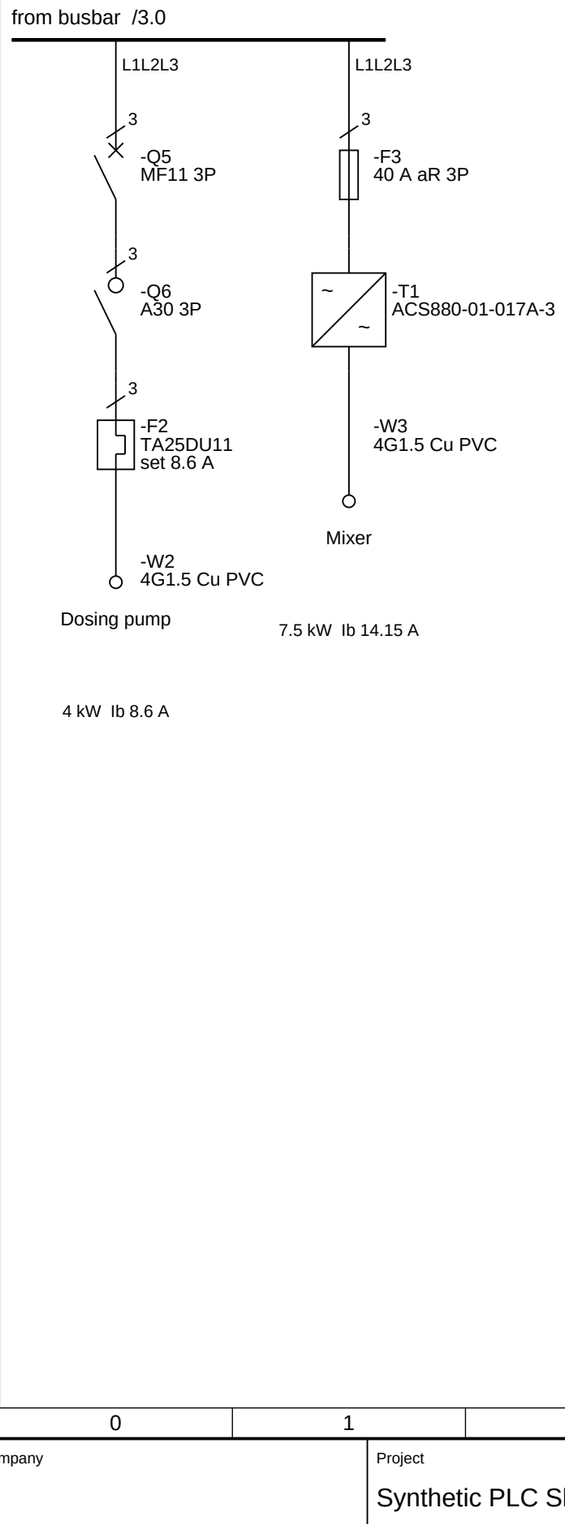

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No.	Note
1	Group 1 breaker raised to 16 A, 1.6 x the 10 A breaker after it, for discrimination.
2	Dosing pump: Ir 8.6 A, the typical Ir of a 4 kW motor in ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3; check it against the nameplate.
3	Dosing pump: dol starter, Type 2 coordination, ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3 Table 3: 400 V 50 kA DOL Normal Type 2.
4	Mixer: Ir 14.15 A from 7.5 kW shaft power at cos phi 0.85, efficiency 0.9; check it against the nameplate.
5	Mixer: ACS880-01-017A-3 for normal duty at 30 °C, behind 40 A aR fuses (ABB ACS880-01 hardware manual (3AUA0000078093)). Use a screened, symmetrical motor cable, earthed 360° at both ends.
6	Mixer: the fuses need at least 140 A prospective short-circuit current at the board to operate fast enough.
7	Incomer raised to 32 A, 1.6 x the 16 A breaker after it, for discrimination.
8	Breakers are rated for at least 10 kA breaking capacity against 10 kA prospective at the board; each RCCB's conditional short-circuit current with the breaker ahead of it must reach it too.
9	Leave 1 spare outgoing ways (20 %).
10	Loads without a power factor were taken at cos phi 0.9.
11	Contactors are rated at least their breaker's current (AC-1); confirm the utilisation category against the catalogue for motor or lamp loads.
12	Each breaker is rated at least 1.6 x the largest breaker after it, so they discriminate on overload. On short circuit, miniature breakers discriminate only up to the upstream breaker's instantaneous trip (about 5 x In for curve C); where total selectivity is required, confirm it from the manufacturer's selectivity tables.
13	3 circuit(s) have no cable length given; their voltage drop is not checked.
14	No Ze is given for this board, so circuits not under a residual current device are not checked for disconnection on an earth fault; enter the measured or declared Ze.
15	Each cable's withstand k^2S^2 is in the cable list; compare it with the selected breaker's let-through I^2t at 10 kA from its maker's curve (ABB handbook §2.4).
16	Cables sized for method B1, 30 °C, 1 grouped circuit(s), 70 °C insulation.
17	Circuit rules are this software's defaults, not confirmed by the company's engineers.
18	Enclosure: none named, since 6 devices have no width (rail_widths_mm in the company settings).

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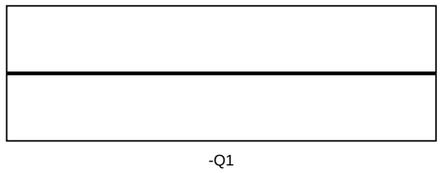
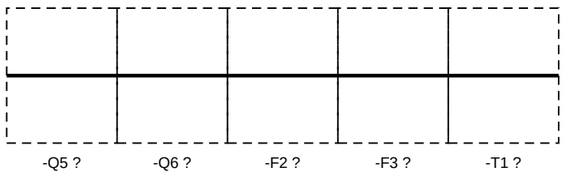
Incomer: 70 mm of rail

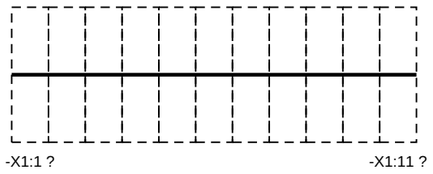


Residual current group: 140 mm of rail; 1 without a width

Busbar: 0 mm of rail; 5 without a width



Terminals: 0 mm of rail; 11 without a width



Dashed: no width is given for the device's kind (rail_widths_mm in the company settings).

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Terminal	Conductor	Circuit	Cable	mm²	Article
-X1:1	L1	Panel lighting	-W1	1.5	8WH1000-0AF00
-X1:2	N	Panel lighting	-W1	1.5	8WH1000-0AF00
-X1:3	PE	Panel lighting	-W1	1.5	8WH1000-0CF07
-X1:4	U	Dosing pump	-W2	1.5	8WH1000-0AF00
-X1:5	V	Dosing pump	-W2	1.5	8WH1000-0AF00
-X1:6	W	Dosing pump	-W2	1.5	8WH1000-0AF00
-X1:7	PE	Dosing pump	-W2	1.5	8WH1000-0CF07
-X1:8	U	Mixer	-W3	1.5	8WH1000-0AG00
-X1:9	V	Mixer	-W3	1.5	8WH1000-0AG00
-X1:10	W	Mixer	-W3	1.5	8WH1000-0AG00
-X1:11	PE	Mixer	-W3	1.5	8WH1000-0CG07

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Cable	From	To	Type	Length
CP-1 -W1	CP-1 -Q3	Panel lighting	3G1.5 Cu PVC	
CP-1 -W2	CP-1 -Q5	Dosing pump	4G1.5 Cu PVC	
CP-1 -W3	CP-1 -F3	Mixer	4G1.5 Cu PVC	

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	Synthetic PLC Skid			**Cable list**	
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Qty	Description	Article	Order number	Devices
1	Circuit-breaker C32 4P 10 kA	not selected		-Q1
1	Circuit-breaker C16 3P 10 kA	not selected		-Q2
1	Residual current circuit-breaker 16 A 30 mA 4P	not selected		-F1
1	Circuit-breaker MF11 3P	ABB T2S160 MF 11		-Q5
1	Contactor A30 3P	ABB A30		-Q6
1	Overload relay TA25DU11 set 8.6 A	ABB TA25DU11		-F2
1	Fuse 40 A aR 3P	Bussmann 170M1313		-F3
1	Variable speed drive ACS880-01-017A-3	ABB ACS880-01-017A-3		-T1
1	Circuit-breaker C10 1P 10 kA	not selected		-Q3
1	Contactor 20 A AC-1 2P	not selected		-Q4
5	Terminal 8WH1	Siemens 8WH1000-0AF00	8WH1000-0AF00	CP-1 -X1
2	PE terminal 8WH1	Siemens 8WH1000-0CF07	8WH1000-0CF07	CP-1 -X1
3	Terminal 8WH1	Siemens 8WH1000-0AG00	8WH1000-0AG00	CP-1 -X1
1	PE terminal 8WH1	Siemens 8WH1000-0CG07	8WH1000-0CG07	CP-1 -X1

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[BLOCKING] CP-1: required terminal M of component k-4b7b52c55b8da510ac34f651 is in no net (unconnected_required_terminal; source: integrity)
[WARNING] CP-1 / c2: Mixer: the cable after a drive is protected by the drive's motor overload setting, which is not checked here. (integrity_drive_cable_unchecked; source: policy; engineering confirmation required)
[WARNING] CP-1 / c1: Dosing pump: Ir 8.6 A, the typical Ir of a 4 kW motor in ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3; check it against the nameplate. (motor_current_table; source: policy; engineering confirmation required)
[WARNING] CP-1 / c2: Mixer: Ir 14.15 A from 7.5 kW shaft power at cos phi 0.85, efficiency 0.9; check it against the nameplate. (motor_current_formula; source: policy; engineering confirmation required)
[WARNING] CP-1 / c2: Mixer: the fuses need at least 140 A prospective short-circuit current at the board to operate fast enough. (drive_fuse_fault_level; source: policy; engineering confirmation required)
[WARNING] CP-1: Loads without a power factor were taken at cos phi 0.9. (default_power_factor; source: policy; engineering confirmation required)
[WARNING] CP-1: Contactors are rated at least their breaker's current (AC-1); confirm the utilisation category against the catalogue for motor or lamp loads. (contactor_ac1; source: policy; engineering confirmation required)
[WARNING] CP-1: 3 circuit(s) have no cable length given; their voltage drop is not checked. (voltage_drop_unchecked; source: policy; engineering confirmation required)
[WARNING] CP-1: No Ze is given for this board, so circuits not under a residual current device are not checked for disconnection on an earth fault; enter the measured or declared Ze. (earth_fault_no_ze; source: policy; engineering confirmation required)
[WARNING] CP-1: Each cable's withstand k²S² is in the cable list; compare it with the selected breaker's let-through I²t at 10 kA from its maker's curve (ABB handbook §2.4). (short_circuit_withstand; source: policy; engineering confirmation required)
[WARNING] CP-1: Circuit rules are this software's defaults, not confirmed by the company's engineers. (rules_unconfirmed; source: policy; engineering confirmation required)
[WARNING] CP-1: Enclosure: none named, since 6 devices have no width (rail_widths_mm in the company settings). (enclosure_unknown_widths; source: policy; engineering confirmation required)
[WARNING] : uses pattern tenant/synth-di-marshal-4ch/v1, which no qualified engineer has reviewed; it may not be executed (pattern_unreviewed; source: integrity; engineering confirmation required)
[WARNING] CP-1: uses template global/legacy-residual_current_…, which no qualified engineer has reviewed; it may not be executed (template_unreviewed; source: integrity; engineering confirmation required)
[WARNING] CP-1: uses template global/legacy-circuit_breaker/v1, which no qualified engineer has reviewed; it may not be executed (template_unreviewed; source: integrity; engineering confirmation required)
[WARNING] CP-1: uses template global/legacy-fuse/v1, which no qualified engineer has reviewed; it may not be executed (template_unreviewed; source: integrity; engineering confirmation required)
[WARNING] CP-1: uses template global/legacy-circuit_breaker/v1, which no qualified engineer has reviewed; it may not be executed (template_unreviewed; source: integrity; engineering confirmation required)
[WARNING] CP-1: uses template global/legacy-contactor/v1, which no qualified engineer has reviewed; it may not be executed (template_unreviewed; source: integrity; engineering confirmation required)
[WARNING] CP-1: uses template global/legacy-circuit_breaker/v1, which no qualified engineer has reviewed; it may not be executed (template_unreviewed; source: integrity; engineering confirmation required)
[WARNING] CP-1: uses template global/legacy-drive/v1, which no qualified engineer has reviewed; it may not be executed (template_unreviewed; source: integrity; engineering confirmation required)
[WARNING] CP-1: uses template global/legacy-contactor/v1, which no qualified engineer has reviewed; it may not be executed (template_unreviewed; source: integrity; engineering confirmation required)
[WARNING] CP-1: uses template global/legacy-circuit_breaker/v1, which no qualified engineer has reviewed; it may not be executed (template_unreviewed; source: integrity; engineering confirmation required)
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[WARNING] CP-1: uses template global/legacy-circuit_breaker/v1, which no qualified engineer has reviewed; it may not be executed (template_unreviewed; source: integrity; engineering confirmation required)
[WARNING] CP-1: uses template global/legacy-overload_relay/v1, which no qualified engineer has reviewed; it may not be executed (template_unreviewed; source: integrity; engineering confirmation required)
[WARNING] CP-1: uses template tenant/synth-io-4di/v1, which no qualified engineer has reviewed; it may not be executed (template_unreviewed; source: integrity; engineering confirmation required)
[WARNING] CP-1: uses template tenant/synth-strip-2level/v1, which no qualified engineer has reviewed; it may not be executed (template_unreviewed; source: integrity; engineering confirmation required)
[WARNING] CP-1: wire w-090a29bf8baceddfa7e04d5d has no colour; no rule sets it yet (wire_attribute_unset; source: integrity; engineering confirmation required)
[WARNING] CP-1: wire w-0a4d3a8edfaf2ac3ff1979e6 has no colour; no rule sets it yet (wire_attribute_unset; source: integrity; engineering confirmation required)
[WARNING] CP-1: wire w-43f67b605bce0b8e202ab0d6 has no colour; no rule sets it yet (wire_attribute_unset; source: integrity; engineering confirmation required)
[WARNING] CP-1: wire w-45ec8f2485916680d1cec5bf has no colour; no rule sets it yet (wire_attribute_unset; source: integrity; engineering confirmation required)
[WARNING] CP-1: wire w-884bd1aee40da6e8f9bf32b6 has no colour; no rule sets it yet (wire_attribute_unset; source: integrity; engineering confirmation required)
[WARNING] CP-1: wire w-d31eae56b7fa011ddd71dacf has no colour; no rule sets it yet (wire_attribute_unset; source: integrity; engineering confirmation required)
[WARNING] CP-1: wire w-d3afe8e7d94657f9ee29f95d has no colour; no rule sets it yet (wire_attribute_unset; source: integrity; engineering confirmation required)
[WARNING] CP-1: wire w-090a29bf8baceddfa7e04d5d has no cross_section_mm2; no rule sets it yet (wire_attribute_unset; source: integrity; engineering confirmation required)
[WARNING] CP-1: wire w-0a4d3a8edfaf2ac3ff1979e6 has no cross_section_mm2; no rule sets it yet (wire_attribute_unset; source: integrity; engineering confirmation required)
[WARNING] CP-1: wire w-43f67b605bce0b8e202ab0d6 has no cross_section_mm2; no rule sets it yet (wire_attribute_unset; source: integrity; engineering confirmation required)
[WARNING] CP-1: wire w-45ec8f2485916680d1cec5bf has no cross_section_mm2; no rule sets it yet (wire_attribute_unset; source: integrity; engineering confirmation required)
[WARNING] CP-1: wire w-884bd1aee40da6e8f9bf32b6 has no cross_section_mm2; no rule sets it yet (wire_attribute_unset; source: integrity; engineering confirmation required)
[WARNING] CP-1: wire w-d31eae56b7fa011ddd71dacf has no cross_section_mm2; no rule sets it yet (wire_attribute_unset; source: integrity; engineering confirmation required)
[WARNING] CP-1: wire w-d3afe8e7d94657f9ee29f95d has no cross_section_mm2; no rule sets it yet (wire_attribute_unset; source: integrity; engineering confirmation required)
[WARNING] CP-1: wire w-090a29bf8baceddfa7e04d5d has no number; no rule sets it yet (wire_attribute_unset; source: integrity; engineering confirmation required)
[WARNING] CP-1: wire w-0a4d3a8edfaf2ac3ff1979e6 has no number; no rule sets it yet (wire_attribute_unset; source: integrity; engineering confirmation required)
[WARNING] CP-1: wire w-43f67b605bce0b8e202ab0d6 has no number; no rule sets it yet (wire_attribute_unset; source: integrity; engineering confirmation required)
[WARNING] CP-1: wire w-45ec8f2485916680d1cec5bf has no number; no rule sets it yet (wire_attribute_unset; source: integrity; engineering confirmation required)
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[WARNING] CP-1: wire w-d31eae56b7fa011ddd71dacf has no number; no rule sets it yet (wire_attribute_unset; source: integrity; engineering confirmation required)
[WARNING] CP-1: wire w-d3afe8e7d94657f9ee29f95d has no number; no rule sets it yet (wire_attribute_unset; source: integrity; engineering confirmation required)

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[INFO] CP-1: Group 1 breaker raised to 16 A, 1.6 x the 10 A breaker after it, for discrimination. (discrimination_group_raised; source: policy)
[INFO] CP-1 / c1: Dosing pump: dol starter, Type 2 coordination, ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3 Table 3: 400 V 50 kA DOL Normal Type 2. (starter_table; source: policy)
[INFO] CP-1 / c2: Mixer: ACS880-01-017A-3 for normal duty at 30 °C, behind 40 A aR fuses (ABB ACS880-01 hardware manual (3AUA0000078093)). Use a screened, symmetrical motor cable, earthed 360° at both ends. (drive_selected; source: policy)
[INFO] CP-1: Incomer raised to 32 A, 1.6 x the 16 A breaker after it, for discrimination. (discrimination_incomer_raised; source: policy)
[INFO] CP-1: Breakers are rated for at least 10 kA breaking capacity against 10 kA prospective at the board; each RCCB's conditional short-circuit current with the breaker ahead of it must reach it too. (breaking_capacity; source: policy)
[INFO] CP-1: Leave 1 spare outgoing ways (20 %). (spare_ways; source: policy)
[INFO] CP-1: Each breaker is rated at least 1.6 x the largest breaker after it, so they discriminate on overload. On short circuit, miniature breakers discriminate only up to the upstream breaker's instantaneous trip (about 5 x In for curve C); where total selectivity is required, confirm it from the manufacturer's selectivity tables. (discrimination; source: policy)
[INFO] CP-1: Cables sized for method B1, 30 °C, 1 grouped circuit(s), 70 °C insulation. (cable_conditions; source: policy)